

DATABASE MANAGEMENT SYSTEM

Module 1: Introduction to Database Systems

Complete Exam Notes with Question & Answers

1. Database System Applications

Q1. What is a database? Where is it used?

A database is an organized collection of related data stored and accessed electronically. A Database Management System (DBMS) is software that manages this data.

Common Applications:

- Banking – managing accounts, transactions, loans
- Airlines – flight bookings, schedules, seat reservations
- E-Commerce – products, orders, customers, payments
- Universities – student records, courses, results, fees
- Healthcare – patient records, prescriptions, appointments
- Social Media – user profiles, posts, connections

Key Point: Any system that stores and retrieves structured data uses a DBMS.

2. Database System vs File System

Q2. What are the disadvantages of file system? How does DBMS solve them?

Before DBMS, data was stored in plain files (like .txt or .csv). This caused many problems:

File System Problems	DBMS Solution
Data Redundancy – same data stored in multiple files	No Redundancy – single source of truth
Data Inconsistency – updates in one file don't reflect elsewhere	Consistency – all changes are synchronized
No Security – anyone can open and edit files	Access Control – roles and permissions enforced
No Concurrent Access – two users can't edit at the same time	Concurrency Control – multiple users handled safely
Data Loss on Crash – no recovery mechanism	Recovery – automatic backup and rollback support
No Standard Query – custom code needed to access data	SQL – standard language for all queries

3. Data Abstraction

Q3. What is data abstraction? Explain the three levels.

Data abstraction means hiding unnecessary details from users. DBMS provides 3 levels of abstraction:

Level 1 – Physical Level (Lowest)

Describes HOW data is actually stored on disk — as files, blocks, indexes, etc.

Example: Data stored as B-tree index on hard disk.

Level 2 – Logical Level (Conceptual)

Describes WHAT data is stored and the relationships between data. Uses tables, columns, and data types.

Example: Student(id INTEGER, name VARCHAR, grade CHAR)

Level 3 – View Level (Highest)

Describes WHAT a particular user sees — different users see different parts of the database.

Example: A student only sees their own grades, not other students'.

Important: This separation is called Data Independence — changing one level does not affect others.

4. Instances and Schemas

Q4. Differentiate between schema and instance.

Schema	Instance
The overall design/structure of the database	The actual data stored at a particular moment
Like a blueprint of a building	Like the building after it is built
Defined using DDL (Data Definition Language)	Changes frequently as data is added/deleted
Rarely changes	Changes every day
Example: Student(id, name, dept)	Example: {1001, Ravi, CS}, {1002, Priya, EC}

5. Relational Model & Other Data Models

Q5. What is the relational model? Name other models.

A data model defines how data is structured, stored, and manipulated in a DBMS.

Relational Model (Most Popular)

- Data is stored in tables (called relations)
- Each table has rows (tuples) and columns (attributes)
- Tables are linked using keys (Primary Key and Foreign Key)
- Uses SQL as query language
- Examples: MySQL, Oracle, PostgreSQL, MS SQL Server

Other Data Models

- Hierarchical Model – data organized as a tree (parent-child). Example: IBM IMS
 - Network Model – data organized as a graph (many-to-many links). Example: IDS
 - Object-Oriented Model – data stored as objects, like in OOP languages
 - NoSQL Models – document, key-value, graph databases. Example: MongoDB, Redis
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6. Database Languages

Q6. What are the database languages? Explain DDL and DML.

DBMS provides special languages to interact with the database:

DDL – Data Definition Language

Used to define the structure (schema) of the database.

- CREATE TABLE – creates a new table
- ALTER TABLE – modifies an existing table
- DROP TABLE – deletes a table

Example: CREATE TABLE Student (id INT, name VARCHAR(50), grade CHAR(2));

DML – Data Manipulation Language

Used to insert, update, delete, and retrieve data from the database.

- SELECT – retrieves data
- INSERT INTO – adds new rows
- UPDATE – modifies existing data
- DELETE – removes rows

Example: SELECT name, grade FROM Student WHERE grade = 'A';

DCL – Data Control Language

Controls access and permissions.

- GRANT – gives permission to a user
 - REVOKE – takes back permission
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7. Database Users and Administrator

Q7. Who are the different types of database users?

1. Naive / End Users

Regular users who use applications without knowing SQL. Example: ATM users, students checking results online.

2. Application Programmers

Developers who write programs (in Java, Python, etc.) that interact with the database using SQL.

3. Sophisticated Users (Analysts)

Users who write SQL queries directly to extract reports and analyze data.

4. DBA – Database Administrator

The most important user. The DBA controls and manages the entire database system.

Responsibilities of DBA:

- Defines the schema (structure) of the database
 - Grants and revokes access permissions
 - Monitors database performance
 - Takes backups and handles recovery
 - Tunes the database for better speed
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8. Database System Structure

Q8. Explain the overall structure of a DBMS.

A DBMS has two main internal components:

1. Query Processor

Handles all user queries (SQL statements). It includes the Parser, Translator, and Optimiser.

2. Storage Manager

Manages how data is stored on disk and handles reading/writing. Includes Buffer Manager, File Manager, Transaction Manager, and Authorisation Manager.

Flow:

User writes SQL → Query Processor parses and optimises it → Storage Manager fetches data from disk → Result returned to user.

9. Storage Manager

Q9. What is a storage manager? Explain its components.

The Storage Manager acts as a bridge between the high-level SQL queries and the low-level data stored on disk.

Its 4 main components are:

1. File Manager

Manages the physical storage files and the allocation of space on disk.

2. Buffer Manager

Loads data from disk into main memory (RAM) for faster access. Manages the buffer pool.

3. Transaction Manager

Ensures that all database transactions follow ACID properties:

- Atomicity – all or nothing (transaction fully completes or fully rolls back)
- Consistency – database remains in a valid state after every transaction
- Isolation – concurrent transactions don't interfere with each other
- Durability – once committed, data is permanently saved

4. Authorisation and Integrity Manager

Checks that users have permission to perform operations and that data satisfies integrity constraints.

10. The Query Processor

Q10. How does the query processor work? Explain its components.

The Query Processor takes a user's SQL query and converts it into actual data retrieval from disk.

Step-by-Step Process:

- Step 1 – Parser: Checks the SQL for syntax errors and validates table/column names
- Step 2 – Translator: Converts SQL into relational algebra expressions
- Step 3 – Optimiser: Finds the most efficient execution plan (chooses best indexes, join order, etc.)
- Step 4 – Evaluator / Executor: Runs the execution plan and returns the result to the user

Goal of Optimiser: To execute the query as FAST as possible using minimum resources.

11. Two-Tier and Three-Tier Architecture

Q11. What is two-tier and three-tier architecture in DBMS?

These describe how a client application connects to the database server.

Two-Tier Architecture

The client application talks DIRECTLY to the database server.

- Client sends SQL queries directly to the DBMS
- Simple to build, but less secure
- Good for small applications
- Example: A desktop application like MS Access connected to a local MySQL database

Three-Tier Architecture

An application server sits BETWEEN the client and the database.

- Tier 1 – Client (browser/mobile app): Only displays the user interface
- Tier 2 – Application Server: Contains business logic, processes requests
- Tier 3 – Database Server: Stores and retrieves data

Two-Tier	Three-Tier
Client talks directly to DB	Application server sits in the middle
Simpler to develop	More complex to develop
Less secure	More secure — DB is hidden from client
Less scalable	Highly scalable
Good for small/internal apps	Good for web and enterprise apps
Example: Desktop apps	Example: Web apps, banking portals

Quick Revision — Key Definitions

Term	Definition
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DBMS	Software that manages creation, storage, and retrieval of data in a database
Schema	The logical structure/design of the database (blueprint)
Instance	The actual data stored in the database at a given point in time
DDL	Data Definition Language — used to define database structure (CREATE, ALTER, DROP)
DML	Data Manipulation Language — used to manipulate data (SELECT, INSERT, UPDATE, DELETE)
DBA	Database Administrator — person who manages and controls the DBMS
Transaction	A unit of work that must be executed completely or not at all (ACID)
Query Optimiser	Component that finds the fastest way to execute a SQL query
Buffer Manager	Manages caching of disk data in RAM for faster access
Data Abstraction	Hiding internal complexity from users through 3 levels: Physical, Logical, View
Two-Tier	Architecture where client connects directly to the database
Three-Tier	Architecture with client, application server, and database server layers

— End of Module 1 Notes —